

Separable Equations

ANSWER KEY



Order and verifying solutions

- 1 State the order of the differential equation $y'' + 3y' - 2y = \sin x$.
Second order (the highest derivative appearing is y'').
- 2 State the order of $\left(\frac{dy}{dx}\right)^3 + xy = 0$, and note that order depends only on the highest derivative, not on any power of it.
First order — the equation involves only y' (cubed), no higher derivative.
- 3 Verify that $y = e^{2x}$ is a solution of $y' - 2y = 0$.
 $y' = 2e^{2x}$, so $y' - 2y = 2e^{2x} - 2e^{2x} = 0$. Yes, it is a solution.
- 4 Verify that $y = \frac{1}{x^2}$ is a solution of $y' + y^2 = 0$ for $x \neq 0$.
 $y' = -\frac{1}{x^2}$ and $y^2 = \frac{1}{x^2}$, so $y' + y^2 = -\frac{1}{x^2} + \frac{1}{x^2} = 0$. Yes, it is a solution.

Recognizing and solving separable equations

- 5 Solve $\frac{dy}{dx} = xy$.
Separate: $\frac{dy}{y} = x dx$. Integrate: $\ln|y| = \frac{x^2}{2} + C$. So $y = Ae^{x^2/2}$.
- 6 Solve $\frac{dy}{dx} = \frac{y^2}{x}$ for $x > 0$.
Separate: $\frac{dy}{y^2} = \frac{dx}{x}$. Integrate: $-\frac{1}{y} = \ln x + C$. So $y = \frac{-1}{\ln x + C}$.
- 7 Solve $\frac{dy}{dx} = \frac{x+1}{y}$.
Separate: $y dy = (x+1) dx$. Integrate: $\frac{y^2}{2} = \frac{x^2}{2} + x + C$. So $y^2 = x^2 + 2x + C_1$ (renaming $2C$ as C_1).
- 8 Solve $\frac{dy}{dx} = 6x^2 y^2$.
Separate: $\frac{dy}{y^2} = 6x^2 dx$. Integrate: $-\frac{1}{y} = 2x^3 + C$. So $y = \frac{-1}{2x^3 + C}$.

Equilibrium solutions and qualitative behavior

- 9 Find the equilibrium (constant) solutions of $\frac{dy}{dx} = y(3-y)$.
Set $y(3-y) = 0$: the equilibrium solutions are $y = 0$ and $y = 3$.
- 10 For $\frac{dy}{dx} = y(3-y)$, determine whether $y = 0$ and $y = 3$ are stable or unstable by checking the sign of $\frac{dy}{dx}$ just above and below each.
For $0 < y < 3$: $y(3-y) > 0$, so solutions increase away from $y = 0$ and toward $y = 3$. For $y > 3$:
 $y(3-y) < 0$, so solutions decrease back toward $y = 3$. Hence $y = 0$ is unstable and $y = 3$ is stable.

- 11 Find the equilibrium solutions of $\frac{dy}{dx} = (y - 2)(y + 1)$ and classify each as stable or unstable.
Equilibria: $y = 2$ and $y = -1$. For $-1 < y < 2$, $(y - 2)(y + 1) < 0$, so solutions decrease toward $y = -1$: $y = -1$ is stable. For $y > 2$, the product is positive, pushing solutions away: $y = 2$ is unstable.

General versus particular solutions

- 12 Find the general solution of $\frac{dy}{dx} = 2x$, then find the particular solution satisfying $y(1) = 3$.
General solution: $y = x^2 + C$ (this one is already separated, with $\frac{dy}{dx}$ depending only on x). Since $y(1) = 3$: $1 + C = 3 \Rightarrow C = 2$. Particular solution: $y = x^2 + 2$.
- 13 Explain, in terms of the arbitrary constant C , why a differential equation typically has infinitely many solutions but an initial value problem typically has exactly one.
Integrating introduces one arbitrary constant C , so the general solution is really a one-parameter family of curves, all satisfying the ODE. An initial condition $y(x_0) = y_0$ picks out the single curve in that family passing through (x_0, y_0) , which (under standard existence/uniqueness conditions) fixes C to exactly one value.

Initial value problems

- 14 Solve $\frac{dy}{y} = 3x^2 dx$, $y(0) = 2$.
 $\frac{dy}{y} = 3x^2 dx \Rightarrow \ln|y| = x^3 + C \Rightarrow y = Ae^{x^3}$. Since $y(0) = 2$, $A = 2$: $y = 2e^{x^3}$.
- 15 Solve $\frac{dy}{y} = -2x dx$, $y(0) = 5$.
 $\frac{dy}{y} = -2x dx \Rightarrow \ln|y| = -x^2 + C \Rightarrow y = Ae^{-x^2}$. Since $y(0) = 5$, $A = 5$: $y = 5e^{-x^2}$.
- 16 Solve $\frac{dy}{y} = 4x^3 dx$, $y(0) = -3$.
 $\frac{dy}{y} = 4x^3 dx \Rightarrow \ln|y| = x^4 + C \Rightarrow y = Ae^{x^4}$. Since $y(0) = -3$, $A = -3$: $y = -3e^{x^4}$.
- 17 Solve $y' = -\frac{1}{2}y$, $y(0) = 8$.
 $\frac{dy}{y} = -\frac{1}{2} dx \Rightarrow \ln|y| = -\frac{x}{2} + C \Rightarrow y = Ae^{-x/2}$. Since $y(0) = 8$, $A = 8$: $y = 8e^{-x/2}$.

Application: radioactive decay and population growth

- 18 A radioactive substance decays according to $\frac{dA}{dt} = -kA$. If $A(0) = 100$ g and $A(5) = 50$ g, find k .
 $A = 100e^{-kt}$. Then $50 = 100e^{-5k} \Rightarrow e^{-5k} = \frac{1}{2} \Rightarrow -5k = \ln \frac{1}{2} \Rightarrow k = \frac{\ln 2}{5} \approx 0.1386$ per unit time.
- 19 A population grows according to $\frac{dP}{dt} = 0.03P$, with $P(0) = 200$. Find $P(10)$.
 $P(t) = 200e^{0.03t}$, so $P(10) = 200e^{0.3} \approx 269.97$.

- 20 A bank account earns continuously-compounded interest, $\frac{dA}{dt} = 0.05A$, with $A(0) = 1000$. Find $A(20)$ and the doubling time.
 $A(t) = 1000e^{0.05t}$, so $A(20) = 1000e^1 \approx 2718.28$. Doubling time solves $e^{0.05t} = 2 \Rightarrow t = \frac{\ln 2}{0.05} \approx 13.86$ years.

Application: Newton's Law of Cooling

- 21 Newton's Law of Cooling gives $\frac{dT}{dt} = -k(T - T_s)$, where T_s is the surroundings' temperature. Separate variables and integrate to find the general solution.
 Separate: $\frac{dT}{T - T_s} = -k dt$. Integrate: $\ln|T - T_s| = -kt + C$. So $T = T_s + Ae^{-kt}$.
- 22 A cup of coffee at 90°C is left in a 20°C room. If $T(10) = 60^\circ \text{C}$, find k , then find $T(20)$.
 $T = 20 + Ae^{-kt}$, $T(10) = 60 \Rightarrow 1 - 70A = 70e^{-10k}$, $\frac{T(10) - 20}{10 - 20} = \frac{60 - 20}{10 - 20} = 70e^{-10k} \Rightarrow k = \frac{1}{10} \ln \frac{7}{4} \approx 0.056$. Then
 $T(20) = 20 + 70e^{-20k} = 20 + 70\left(\frac{4}{7}\right)^2 = 20 + 70 \cdot \frac{16}{49} \approx 42.86^\circ \text{C}$.

Further practice: separable equations

- 23 Solve $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$.
 Separate: $\frac{dy}{1+y^2} = \frac{dx}{1+x^2}$. Integrate: $\arctan y = \arctan x + C$.
- 24 Solve $y' = y \cos x$, $y(0) = 1$.
 $\frac{dy}{y} = \cos x dx \Rightarrow \ln|y| = \sin x + C \Rightarrow y = Ae^{\sin x}$. Since $y(0) = 1$, $A = 1$: $y = e^{\sin x}$.
- 25 Solve $\frac{dy}{dx} = e^{x-y}$.
 Rewrite $e^{x-y} = e^x e^{-y}$, so $e^y dy = e^x dx$. Integrate: $e^y = e^x + C$, i.e., $y = \ln(e^x + C)$.
- 26 Solve $x \frac{dy}{dx} = y \ln y$ for $x, y > 0$, $y(1) = e$.
 Separate: $\frac{dy}{y \ln y} = \frac{dx}{x}$. Integrate (left side is $\int \frac{du}{u}$ with $u = \ln y$): $\ln|\ln y| = \ln x + C \Rightarrow \ln y = Ax$. Since $y(1) = e$: $\ln e = A \Rightarrow A = 1$. So $\ln y = x$, i.e., $y = e^x$.

Mixed review

- 27 Determine whether $\frac{dy}{dx} = x + y$ is separable. Justify your answer.
 No. The right side $x + y$ cannot be factored as (a function of x) $\times$ (a function of y), since it is a sum, not a product.
- 28 Solve $\frac{dy}{dx} = \frac{\cos x}{y}$, $y(0) = -2$.
 Separate: $y dy = \cos x dx$. Integrate: $\frac{y^2}{2} = \sin x + C$. Since $y(0) = -2$: $2 = 0 + C \Rightarrow C = 2$, so $y^2 = 2\sin x + 4$. Since $y(0) = -2 < 0$, take the negative branch: $y = -\sqrt{2\sin x + 4}$.

- 29** A tank of water drains according to Torricelli's Law, $\frac{dh}{dt} = -kh^{1/2}$, where h is the water height. Separate variables and integrate.
 Separate: $h^{-1/2} dh = -k dt$. Integrate: $2h^{1/2} = -kt + C$, so $h = \left(\frac{C - kt}{2}\right)^2$.
- 30** For the model $\frac{dy}{dx} = \frac{2x}{y}$ with $y(0) = 3$. find the particular solution and state its domain.
 Separate: $y dy = 2x dx \Rightarrow \frac{y^2}{2} = x^2 + C$. Since $y(0) = 3$: $\frac{9}{2} = C$, so $y^2 = 4x^2 + 9$, and since $y(0) = 3 > 0$:
 $y = \sqrt{4x^2 + 9}$, defined for all real x .